

FIFA 21 PLAYER MARKET & PERFORMANCE ANALYSIS

Exploratory Data Analysis (EDA) Report -

QUESTION 1: Which factors have the greatest impact on a player's market value?

- **Objective:** Identify the key player attributes that most strongly influence a player's market value.
- **Methodology:** Analyze the relationship between market value and player features such as overall rating, potential, age, wage, and performance statistics using exploratory data analysis and correlation techniques.

SOLUTION:

```
In [3]: # 1. Isolate only the numeric columns so the math doesn't crash on text names
numeric_df = df.select_dtypes(include=['int64', 'float64'])

# 2. Calculate how every numeric column correlates specifically with Market 'Value'
value_correlations = numeric_df.corr()['Value'].sort_values(ascending=False)

# 3. Print the top drivers of market value
print("--- TOP 10 STATS THAT INCREASE MARKET VALUE ---")
print(value_correlations.head(11)) # Using 11 because 'Value' correlating with itself (1.0) will be first

print("\n--- STATS THAT DECREASE MARKET VALUE ---")
print(value_correlations.tail(5))
```

OUTPUT:

```
--- TOP 10 STATS THAT INCREASE MARKET VALUE ---
Value          1.000000
Release Clause  0.966440
Wage            0.814870
Best_Overall    0.563253
Overall         0.552893
International_Reputation  0.533569
Potential       0.528200
Reactions       0.489513
Base Stats      0.462458
PAS             0.394543
DRI             0.393347
Name: Value, dtype: float64

--- STATS THAT DECREASE MARKET VALUE ---
Goalkeeping    -0.011616
GK Diving      -0.012163
GK Kicking     -0.012592
Potential_Growth -0.110061
ID             -0.131001
Name: Value, dtype: float64
```

- **Business Insight:** Overall rating, potential, and age are typically the strongest drivers of market value, helping clubs make smarter recruitment, pricing, and investment decisions.

QUESTION 2: Does a higher overall rating always result in a higher market value?

- **Objective:** Determine whether a higher overall rating consistently leads to a higher player market value.
- **Methodology:** Compare players' overall ratings with their market values using scatter plots and correlation analysis to identify trends and exceptions.

SOLUTION:

```
In [4]: # 1. Group players by Age and calculate the average stats for each year
age_analysis = df.groupby('Age')[['Overall', 'Potential', 'Value', 'Wage']].mean()

# 2. Convert Value and Wage to clean Millions/Thousands for easier reading
age_analysis['Value (Millions €)'] = (age_analysis['Value'] / 1000000).round(2)
age_analysis['Wage (Thousands €)'] = (age_analysis['Wage'] / 1000).round(2)
age_analysis['Overall'] = age_analysis['Overall'].round(1)

# 3. Display the career arc from age 18 to 35
print("--- PLAYER LIFECYCLE (AVERAGE STATS BY AGE) ---")
print(age_analysis[['Overall', 'Value (Millions €)', 'Wage (Thousands €)']].loc[18:35])
```

OUTPUT:

```
--- PLAYER LIFECYCLE (AVERAGE STATS BY AGE) ---
      Overall  Value (Millions €)  Wage (Thousands €)
Age
18      57.4          0.73         1.33
19      58.4          1.08         2.63
20      61.5          1.88         4.54
21      62.5          2.49         4.81
22      64.4          2.92         7.25
23      65.3          3.37         8.40
24      67.1          3.63         9.23
25      67.3          3.78        10.66
26      67.8          3.74        11.89
27      68.3          3.79        12.17
28      68.9          3.83        14.38
29      68.8          3.63        13.64
30      68.8          3.16        12.10
31      69.1          2.97        12.39
32      69.5          2.94        12.50
33      69.8          2.71        14.82
34      69.3          1.54        12.03
35      69.1          1.27         9.04
```

- **Business Insight:** Higher overall ratings generally increase market value, but factors such as age, potential, contract length, and popularity also significantly influence a player's valuation.

QUESTION 3: How does player age influence market value?

- **Objective:** Analyze the relationship between player age and market value to identify the age range with the highest valuation.
- **Methodology:** Examine player ages against market values using descriptive statistics and visualizations to identify age-based valuation patterns.

SOLUTION:

```
In [4]: # 1. Group players by Age and calculate the average stats for each year
age_analysis = df.groupby('Age')[['Overall', 'Potential', 'Value', 'Wage']].mean()

# 2. Convert Value and Wage to clean Millions/Thousands for easier reading
age_analysis['Value (Millions €)'] = (age_analysis['Value'] / 1000000).round(2)
age_analysis['Wage (Thousands €)'] = (age_analysis['Wage'] / 1000).round(2)
age_analysis['Overall'] = age_analysis['Overall'].round(1)

# 3. Display the career arc from age 18 to 35
print("--- PLAYER LIFECYCLE (AVERAGE STATS BY AGE) ---")
print(age_analysis[['Overall', 'Value (Millions €)', 'Wage (Thousands €)']].loc[18:35])
```

OUTPUT:

```
--- PLAYER LIFECYCLE (AVERAGE STATS BY AGE) ---
Overall  Value (Millions €)  Wage (Thousands €)
Age
18      57.4                0.73              1.33
19      58.4                1.08              2.63
20      61.5                1.88              4.54
21      62.5                2.49              4.81
22      64.4                2.92              7.25
23      65.3                3.37              8.40
24      67.1                3.63              9.23
25      67.3                3.78             10.66
26      67.8                3.74             11.89
27      68.3                3.79             12.17
28      68.9                3.83             14.38
29      68.8                3.63             13.64
30      68.8                3.16             12.10
31      69.1                2.97             12.39
32      69.5                2.94             12.50
33      69.8                2.71             14.82
34      69.3                1.54             12.03
35      69.1                1.27              9.04
```

- **Business Insight:** Players in their prime years generally command the highest market values, while younger players gain value through potential and older players typically experience declining valuations.

QUESTION 4: Which young players have the highest future potential?

- **Objective:** Identify young players with the highest potential ratings to highlight future stars and long-term investment opportunities.
- **Methodology:** Filter young players based on age, rank them by potential rating, and compare their overall performance and market value.

SOLUTION:

```
In [5]: # 1. Filter for young, high-growth, and affordable players
moneyball_targets = df[(df['Age'] <= 23) &
                        (df['Potential_Growth'] >= 5) &
                        (df['Value'] < 15000000)]

# 2. Sort by highest Potential, then lowest Wage to find the best ROI
best_targets = moneyball_targets.sort_values(by=['Potential', 'Wage'], ascending=[False, True])

# 3. Display the Top 10 Hidden Gems
print("--- TOP 10 HIDDEN GEMS (HIGH POTENTIAL, LOW COST) ---")
print(best_targets[['Name', 'Age', 'Overall', 'Potential', 'Potential_Growth', 'Value', 'Wage']].head(10))
```

OUTPUT:

```
--- TOP 10 HIDDEN GEMS (HIGH POTENTIAL, LOW COST) ---
      Name  Age  Overall  Potential  Potential_Growth  Value \
2757  T. Almada  19      73         89             16  8000000.0
1664   T. Kubo  19      75         89             14 13500000.0
8116   R. Cherki 16      67         88             21  3200000.0
6235  J. Bellingham 17      69         88             19  3500000.0
6836   F. Wirtz  17      68         88             20  3300000.0
2962   A. Urzi  20      73         88             15  7500000.0
3791   Pedri  17      72         88             16  6000000.0
4233   J. Doku  18      71         88             17  4800000.0
1781   B. Saka  18      75         88             13 13000000.0
1761  Riqui Puig  20      75         88             13 13500000.0

      Wage
2757  8000.0
1664  59000.0
8116   2000.0
6235   3000.0
6836   3000.0
2962  10000.0
3791  10000.0
4233  11000.0
1781  22000.0
1761  77000.0
```

- **Business Insight:** Young players with high potential represent valuable long-term assets, enabling clubs to invest early and maximize future performance and transfer returns.

QUESTION 5: Which players appear undervalued based on their current ratings?

- **Objective:** Identify players whose market values are lower than expected based on their current overall ratings, indicating potential bargains.
- **Methodology:** Compare players' overall ratings with their market values to detect players who have high ratings but relatively low market values.

SOLUTION:

```
In [6]: # 1. Define what a "Bargain" is:
# High skill (75+), Realistic cost (between 0.5M and 5M)
# This removes the "ghost" free agents with 0 value
bargains = df[(df['Overall'] >= 75) &
              (df['Value'] > 500000) &
              (df['Value'] < 5000000)]

# 2. Sort these bargains: Highest Overall first, then Cheapest price
undervalued = bargains.sort_values(by=['Overall', 'Value'], ascending=[False, True])

# 3. Identify Best Value by Position
# We calculate 'Skill_per_Euro' (Overall / Value)
df['Skill_per_Euro'] = df['Overall'] / (df['Value'] + 1)
pos_investment = df.groupby('Best_Overall')['Skill_per_Euro'].mean().sort_values(ascending=False)

# 4. Print the results
print("--- TOP 10 TRUE BARGAIN PLAYERS (75+ Overall, 0.5M - 5M Value) ---")
print(undervalued[['Name', 'Overall', 'Value', 'Wage']].head(10))

print("\n--- BEST INVESTMENT BY POSITION (Highest Skill per Euro) ---")
print(pos_investment.head(10).round(6))
```

OUTPUT:

```
--- TOP 10 TRUE BARGAIN PLAYERS (75+ Overall, 0.5M - 5M Value) ---
   Name  Overall  Value  Wage
208  S. Mandanda    83  4000000.0  28000.0
262   G. Buffon    82  3400000.0  41000.0
362  L. Fabiański    81  2900000.0  26000.0
366    Pepe       81  4600000.0  13000.0
501  I. Akinfeev    80  3600000.0  40000.0
477  B. Ivanović    80  3800000.0  61000.0
679   B. Foster    79  1900000.0  12000.0
683  Pepe Reina    79  1900000.0  30000.0
672   N. Guzmán    79  3000000.0  40000.0
682   G. Ochoa     79  3000000.0  34000.0

--- BEST INVESTMENT BY POSITION (Highest Skill per Euro) ---
Best_Overall
80    4.618502
77    2.604529
79    2.556763
76    2.087953
72    1.901180
69    1.416151
75    1.401906
70    1.281076
73    1.271566
81    1.208959
Name: Skill_per_Euro, dtype: float64
```

- **Business Insight:** Undervalued players offer cost-effective transfer opportunities, enabling clubs to strengthen their squads while maximizing return on investment.
-

QUESTION 6: Which positions provide the best value for investment?

- **Objective:** Determine which playing positions deliver the highest performance relative to their market value for optimal investment decisions.
- **Methodology:** Compare average market value, overall rating, and potential across player positions to evaluate value for money.

SOLUTION:

```
In [6]: # 1. Define what a "Bargain" is:
# High skill (75+), Realistic cost (between 0.5M and 5M)
# This removes the "ghost" free agents with 0 value
bargains = df[(df['Overall'] >= 75) &
              (df['Value'] > 500000) &
              (df['Value'] < 5000000)]

# 2. Sort these bargains: Highest Overall first, then Cheapest price
undervalued = bargains.sort_values(by=['Overall', 'Value'], ascending=[False, True])

# 3. Identify Best Value by Position
# We calculate 'Skill_per_Euro' (Overall / Value)
df['Skill_per_Euro'] = df['Overall'] / (df['Value'] + 1)
pos_investment = df.groupby('Best_Overall')['Skill_per_Euro'].mean().sort_values(ascending=False)

# 4. Print the results
print("--- TOP 10 TRUE BARGAIN PLAYERS (75+ Overall, 0.5M - 5M Value) ---")
print(undervalued[['Name', 'Overall', 'Value', 'Wage']].head(10))

print("\n--- BEST INVESTMENT BY POSITION (Highest Skill per Euro) ---")
print(pos_investment.head(10).round(6))
```

OUTPUT:

```
--- TOP 10 TRUE BARGAIN PLAYERS (75+ Overall, 0.5M - 5M Value) ---
      Name Overall      Value      Wage
208  S. Mandanda      83 4000000.0  28000.0
262   G. Buffon      82 3400000.0  41000.0
362  L. Fabiański      81 2900000.0  26000.0
366      Pepe      81 4600000.0  13000.0
501  I. Akinfeev      80 3600000.0  40000.0
477  B. Ivanović      80 3800000.0  61000.0
679   B. Foster      79 1900000.0  12000.0
683  Pepe Reina      79 1900000.0  30000.0
672   N. Guzmán      79 3000000.0  40000.0
682   G. Ochoa      79 3000000.0  34000.0

--- BEST INVESTMENT BY POSITION (Highest Skill per Euro) ---
Best_Overall
80    4.618502
77    2.604529
79    2.556763
76    2.087953
72    1.901180
69    1.416151
75    1.401906
70    1.281076
73    1.271566
81    1.208959
Name: Skill_per_Euro, dtype: float64
```

- **Business Insight:** Positions with strong overall ratings and potential at relatively lower market values provide the best investment opportunities for clubs seeking cost-effective squad building.

QUESTION 7: Which playing positions receive the highest average wages?

- **Objective:** Identify the playing positions with the highest average wages to understand salary distribution across the squad.
- **Methodology:** Group players by position and calculate the average wage for each position, then rank them from highest to lowest.

SOLUTION:

```
In [7]: # 1. Group by 'Best Position' instead of 'Best Overall'
pos_salary = df.groupby('Best Position')['Wage'].mean().sort_values(ascending=False)

# 2. Correlation between Wage and Overall Performance
wage_performance_corr = df['Wage'].corr(df['Overall'])

# 3. Print the results
print("--- AVERAGE WAGE BY POSITION ---")
print(pos_salary.head(10).round(2))

print(f"\n--- Correlation between Wage and Overall Performance: {wage_performance_corr:.2f} ---")
```

OUTPUT:

```
--- AVERAGE WAGE BY POSITION ---
Best Position
CF      24163.46
LW      16120.70
RW      15166.61
CM      13582.43
CDM      9682.98
ST      9431.68
CAM      9399.43
RB       9096.25
LM       8949.02
LWB      8805.17
Name: Wage, dtype: float64

--- Correlation between Wage and Overall Performance: 0.59 ---
```

- **Business Insight:** Positions with the highest average wages typically represent premium roles where clubs invest more to secure top-performing players and maintain competitive advantage.

QUESTION 8: Is there a relationship between player wages and overall performance?

- **Objective:** Evaluate whether higher-paid players tend to have better overall performance ratings.
- **Methodology:** Analyze the relationship between player wages and overall ratings using correlation analysis and scatter plot visualization.

SOLUTION:

```
In [7]: # 1. Group by 'Best Position' instead of 'Best Overall'
pos_salary = df.groupby('Best Position')['Wage'].mean().sort_values(ascending=False)

# 2. Correlation between Wage and Overall Performance
wage_performance_corr = df['Wage'].corr(df['Overall'])

# 3. Print the results
print("--- AVERAGE WAGE BY POSITION ---")
print(pos_salary.head(10).round(2))

print(f"\n--- Correlation between Wage and Overall Performance: {wage_performance_corr:.2f} ---")
```

OUTPUT:

```
--- AVERAGE WAGE BY POSITION ---
```

```
Best Position
```

```
CF      24163.46
```

```
LW      16120.70
```

```
RW      15166.61
```

```
CM      13582.43
```

```
CDM      9682.98
```

```
ST       9431.68
```

```
CAM      9399.43
```

```
RB       9096.25
```

```
LM       8949.02
```

```
LWB      8805.17
```

```
Name: Wage, dtype: float64
```

```
--- Correlation between Wage and Overall Performance: 0.59 ---
```

- **Business Insight:** Player wages generally increase with overall performance, although factors such as reputation, experience, club stature, and contract terms also influence salaries.

QUESTION 9: Which clubs have the highest average player ratings?

- **Objective:** Identify the clubs with the highest average player ratings to assess overall squad quality.
- **Methodology:** Group players by club, calculate the average overall rating for each club, and rank the clubs from highest to lowest.

SOLUTION:

```
In [8]: # 1. Filter for clubs with at least 15 players to ensure data quality
club_performance = df.groupby('Club').filter(lambda x: len(x) >= 15)

# 2. Calculate average 'Overall' rating per club
club_ranking = club_performance.groupby('Club')['Overall'].mean().sort_values(ascending=False)

# 3. Print the results
print("--- TOP 10 CLUBS BY AVERAGE SQUAD RATING ---")
print(club_ranking.head(10).round(2))
```

OUTPUT:

```
--- TOP 10 CLUBS BY AVERAGE SQUAD RATING ---
Club
FC Bayern München      81.48
Real Madrid             79.77
Inter                  79.35
Napoli                  78.69
Chelsea                 78.18
Manchester United       78.09
Juventus                78.07
Paris Saint-Germain     77.97
Liverpool               77.39
Sevilla FC              77.29
Name: Overall, dtype: float64
```

- **Business Insight:** Clubs with the highest average player ratings typically possess stronger squads, reflecting greater competitiveness, talent depth, and investment in player quality.

Question 10: Which countries produce the highest-rated players?

- **Objective:** Identify the countries that produce the highest-rated football players based on average overall ratings.
- **Methodology:** Group players by nationality, calculate the average overall rating for each country, and rank the countries from highest to lowest.

SOLUTION:

```
In [9]: # Filter for countries with at least 15 players
nat_performance = df.groupby('Nationality').filter(lambda x: len(x) >= 15)
nat_ranking = nat_performance.groupby('Nationality')['Overall'].mean().sort_values(ascending=False)

print("--- TOP 10 NATIONALITIES BY AVERAGE OVERALL RATING ---")
print(nat_ranking.head(10).round(2))
```

OUTPUT:

```
--- TOP 10 NATIONALITIES BY AVERAGE OVERALL RATING ---
Nationality
Egypt          71.40
Brazil         70.95
Algeria        70.62
Ukraine        70.51
Russia         70.35
Czech Republic 70.20
Iran           70.11
Morocco        69.91
Gabon          69.62
Portugal       69.55
Name: Overall, dtype: float64
```

- **Business Insight:** Countries with the highest average player ratings consistently produce elite talent, helping clubs target strong football markets for recruitment and scouting.

QUESTION 11: What are the overall trends in player distribution across positions and nationalities?

- **Objective:** Analyze the overall distribution of players across key attributes such as age, positions, ratings, and nationalities to understand dataset patterns.
- **Methodology:** Use descriptive statistics and visualizations to examine the frequency and distribution of player characteristics across the dataset.

SOLUTION:

```
In [10]: # 1. Top 10 Nationalities by total player count (Volume)
nat_distribution = df['Nationality'].value_counts().head(10)

# 2. Distribution by 'Best Position' (Landscape)
pos_distribution = df['Best Position'].value_counts()

# 3. Print the results
print("--- TOP 10 NATIONALITIES BY TOTAL PLAYER COUNT (Volume) ---")
print(nat_distribution)

print("\n--- PLAYER DISTRIBUTION BY POSITION (Frequency) ---")
print(pos_distribution)
```

OUTPUT:

```
--- TOP 10 NATIONALITIES BY TOTAL PLAYER COUNT (Volume) ---
Nationality
England      1705
Germany      1195
Spain         1065
France        1003
Argentina      943
Brazil         887
Japan          485
Netherlands   438
Italy          387
Sweden         380
Name: count, dtype: int64

--- PLAYER DISTRIBUTION BY POSITION (Frequency) ---
Best Position
CB          3686
ST          2680
CAM         2299
GK          2075
RM          1611
CDM         1445
LB          1086
RB          1079
CM          1047
LM           871
RW           298
RWB          277
LWB          261
LW           186
CF            78
Name: count, dtype: int64
```

- **Business Insight:** Understanding player distribution helps clubs identify talent trends, optimize scouting strategies, and make data-driven recruitment decisions.

QUESTION 12: Are there players whose Release Clause is significantly lower than their actual Market Value?

- **Objective:** Identify players whose release clauses are significantly lower than their market values, indicating potential transfer bargains.
- **Methodology:** Compare each player's release clause with their market value and identify cases where the release clause is substantially lower than the estimated market value.

SOLUTION:

```
In [11]: # 1. Filter: Find players where Release Clause is > 0 and strictly Less than Market Value
inefficient_assets = df[(df['Release Clause'] > 0) & (df['Release Clause'] < df['Value'])]

# 2. Add Quality Filter: Focus only on "Impact Players" (Overall >= 70)
quality_bargains = inefficient_assets[inefficient_assets['Overall'] >= 70]

# 3. Calculate the 'Discount' percentage
quality_bargains['Discount_Pct'] = ((quality_bargains['Value'] -
                                     quality_bargains['Release Clause']) /
                                     quality_bargains['Value']) * 100

# 4. Sort by the highest discount percentage
final_targets = quality_bargains.sort_values(by='Discount_Pct', ascending=False)

# 5. Display the top 10 "Impact" bargain opportunities
print("--- TOP 10 IMPACT PLAYERS WITH UNDERVALUED RELEASE CLAUSES (70+ Overall) ---")
print(final_targets[['Name', 'Overall', 'Value', 'Release Clause', 'Discount_Pct']].head(10))
```

OUTPUT:

```
--- TOP 10 IMPACT PLAYERS WITH UNDERVALUED RELEASE CLAUSES (70+ Overall) ---
   Name  Overall  Value  Release Clause  Discount_Pct
721  G. Montiel    78  27000000.0    20800000.0    22.962963
284   A. Davies    81  53000000.0    41600000.0    21.509434
698  L. Martínez    78  27000000.0    21600000.0    20.000000
719   D. Malen    78  29000000.0    24000000.0    17.241379
5315  L. Licht    70   350000.0     290000.0    17.142857
711   C. Stengs    78  31500000.0    26400000.0    16.190476
69   G. Donnarumma    85  93500000.0    78900000.0    15.614973
402   N. Mukiele    80  40500000.0    35200000.0    13.086420
687    Antony    78  31000000.0    27200000.0    12.258065
746   A. Meret    78  27000000.0    23800000.0    11.851852
```

- **Business Insight:** Players with release clauses below their market values represent attractive transfer opportunities, allowing clubs to acquire high-value talent at a relatively lower cost.

QUESTION 13: Which high-potential young players have contracts expiring soon?

- **Objective:** Identify high-potential players with contracts nearing expiration to uncover valuable and cost-effective transfer opportunities.
- **Methodology:** Filter players with high potential ratings and short remaining contract durations, then analyze their market values and overall ratings.

SOLUTION:

```
In [12]: # 1. Convert 'Contract_Valid_Until' to numeric for easier filtering
# Note: Ensure your data is clean; this assumes years are 4-digit strings
df['Contract_Year'] = df['Contract_Valid_Until'].str[-4:]
df['Contract_Year'] = pd.to_numeric(df['Contract_Year'], errors='coerce')

# 2. Filter criteria:
# - Young players (Age <= 23)
# - High potential (Potential >= 80)
# - Contract expiring this year or next (e.g., 2026 or 2027)
expiring_soon = df[(df['Age'] <= 23) &
                    (df['Potential'] >= 80) &
                    (df['Contract_Year'].isin([2026, 2027]))]

# 3. Sort by Potential (highest first)
top_targets = expiring_soon.sort_values(by='Potential', ascending=False)

# 4. Display the results
print("--- TOP HIGH-POTENTIAL YOUNGSTERS WITH EXPIRING CONTRACTS ---")
print(top_targets[['Name', 'Age', 'Overall', 'Potential', 'Contract_Valid_Until']].head(10))
```

OUTPUT:

```
--- TOP HIGH-POTENTIAL YOUNGSTERS WITH EXPIRING CONTRACTS ---
      Name  Age  Overall  Potential  Contract_Valid_Until
280  João Félix   20      81         93                2026
521  P. Estupiñán  22      79         86                2027
```

- **Business Insight:** High-potential players with expiring contracts offer clubs an opportunity to secure future talent at reduced transfer costs before contract renewal or free agency.

QUESTION 14. Do players with 5-star Skill Moves or 5-star Weak Foot command a higher wage than players with the exact same Overall rating who lack those traits?

- **Objective:** Determine whether players with 5-star Skill Moves or 5-star Weak Foot earn higher wages than players with the same Overall rating who do not possess these traits.
- **Methodology:** Group players by Overall rating, compare the average wages of players with and without 5-star Skill Moves or 5-star Weak Foot, and analyze the wage differences.

SOLUTION:

```
In [13]: # 1. Create a boolean series to flag the 5-star players
df['Has_5Star_Skills'] = (df['Skill_Moves'] == 5)
df['Has_5Star_WeakFoot'] = (df['Weak_Foot'] == 5)

# 2. Now perform the groupby using these new boolean columns
skill_premium = df.groupby(['Overall', 'Has_5Star_Skills'])['Wage'].mean().unstack()
wf_premium = df.groupby(['Overall', 'Has_5Star_WeakFoot'])['Wage'].mean().unstack()

# 3. Print the comparison for high-rated players (Overall 80+)
print("--- WAGE COMPARISON: SKILL MOVES PREMIUM (Overall 80+) ---")
# Renaming columns for clarity (False = Normal, True = 5-Star)
print(skill_premium.loc[80:].rename(columns={False: 'Normal Skill', True: '5-Star Skill'}))

print("\n--- WAGE COMPARISON: WEAK FOOT PREMIUM (Overall 80+) ---")
print(wf_premium.loc[80:].rename(columns={False: 'Normal WF', True: '5-Star WF'}))
```

OUTPUT:

```
--- WAGE COMPARISON: SKILL MOVES PREMIUM (Overall 80+) ---
Has_5Star_Skills  Normal Skill  5-Star Skill
Overall
80               50701.526718   51500.000000
81               54494.382022   53166.666667
82               64067.796610   26000.000000
83               77363.636364   133500.000000
84               89976.190476   110000.000000
85               118593.750000   153000.000000
86               147153.846154   190000.000000
87               167352.941176   144000.000000
88               211000.000000         NaN
89               273333.333333         NaN
90               226000.000000   160000.000000
91               245000.000000   270000.000000
92                 NaN        220000.000000
93               560000.000000         NaN

--- WAGE COMPARISON: WEAK FOOT PREMIUM (Overall 80+) ---
Has_5Star_WeakFoot  Normal WF  5-Star WF
Overall
80               50663.200000   51500.000000
81               53866.666667   64200.000000
82               62559.322034   115000.000000
83               80188.405797   124666.666667
84               92048.780488   57500.000000
85               122361.111111   155000.000000
86               150214.285714         NaN
87               163789.473684   165000.000000
88               200000.000000   310000.000000
89               273333.333333         NaN
90               215000.000000         NaN
91               182500.000000   320000.000000
92               220000.000000         NaN
93               560000.000000         NaN
```

- **Business Insight:** Elite technical abilities such as 5-star Skill Moves and Weak Foot often command wage premiums, even among players with identical Overall ratings, reflecting their added tactical and commercial value.

QUESTION 15. How heavily do Height and Weight correlate with a player's Overall rating for specific positions?

- **Objective:** Analyze how strongly height and weight correlate with Overall rating across different playing positions.
- **Methodology:** Group players by position and calculate the correlation between height, weight, and Overall rating, using statistical analysis and visualizations to compare positional differences.

SOLUTION:

```
In [14]: # 1. Select relevant columns
physical_cols = ['Height', 'Weight', 'Overall', 'Best Position']

# 2. Group by Position and calculate correlation with 'Overall'
# We use a lambda to get correlation for each group
pos_physical_corr = df[physical_cols].groupby('Best Position').apply(
    lambda x: x[['Height', 'Weight']].corrwith(x['Overall'])
)

# 3. Print the results
print("--- PHYSICAL ATTRIBUTE CORRELATION WITH OVERALL BY POSITION ---")
print(pos_physical_corr.sort_values(by='Height', ascending=False))
```

OUTPUT:

```
--- PHYSICAL ATTRIBUTE CORRELATION WITH OVERALL BY POSITION ---
              Height    Weight
Best Position
CF              0.299000  0.354802
GK              0.157773  0.284073
ST              0.152080  0.267480
CB              0.145668  0.291979
CM              0.098011  0.176470
CDM             0.083145  0.196199
CAM             0.041493  0.162519
RWB             0.036506  0.140499
LWB             0.027407  0.126651
RB              0.027355  0.175911
LM              0.014809  0.105121
RM              0.004704  0.155993
LB             -0.012468  0.139915
RW             -0.082719 -0.003450
LW             -0.105043  0.085707
```

- **Business Insight:** The impact of height and weight on Overall rating varies by position, helping clubs identify the physical attributes that contribute most to success in each playing role.

QUESTION 16. Are players who can play multiple positions cheaper or more expensive on average than single-position specialists?

- **Objective:** Compare the average market value of multi-position players with single-position specialists to evaluate positional versatility as a factor in player valuation.
- **Methodology:** Classify players based on the number of positions they can play, calculate the average market value for each group, and compare the results.

SOLUTION:

```
In [15]: # 1. Count how many positions each player can play
df['Num_Positions'] = df['Positions'].apply(lambda x: len(x.split(',')))

# 2. Calculate average Wage and Overall rating by versatility
versatility_analysis = df.groupby('Num_Positions')['Wage'].mean().sort_index()
versatility_rating = df.groupby('Num_Positions')['Overall'].mean().sort_index()

# 3. Print the formatted findings
print("--- VERSATILITY VALUE SUMMARY ---")
for pos_count in versatility_analysis.index:
    avg_overall = versatility_rating[pos_count]
    avg_wage = versatility_analysis[pos_count]

    # Format: X Position(s): Overall Y | Wage Z
    suffix = "Position" if pos_count == 1 else "Positions"
    print(f"{pos_count} {suffix}: Overall {avg_overall:.2f} | Wage {avg_wage:,.0f}")
```

OUTPUT:

```
--- VERSATILITY VALUE SUMMARY ---
1 Position: Overall 64.73 | Wage 8,137
2 Positions: Overall 66.45 | Wage 9,945
3 Positions: Overall 67.75 | Wage 10,660
```

- **Business Insight:** Multi-position players often command higher market values due to their tactical flexibility, making them valuable assets for clubs seeking versatile squad options.

QUESTION 17. Does a high Hits count (popularity/hype) correlate with an inflated market value?

- **Objective:** Determine whether player popularity (Hits count) is associated with higher market values beyond on-field performance.
- **Methodology:** Analyze the correlation between Hits count and market value using statistical measures and scatter plot visualization to identify any relationship.

SOLUTION:

```
# 1. Calculate the correlation coefficient between Hits and Value
hype_corr = df['Hits'].corr(df['Value'])

# 2. To visualize this, let's look at the average Value of "Trending" vs "Non-Trending" players
# We define "Trending" as players with Hits above the median
median_hits = df['Hits'].median()
df['Is_Trending'] = df['Hits'] > median_hits

trending_comparison = df.groupby('Is_Trending')['Value'].mean()

# 3. Print the findings
print(f"--- CORRELATION: HITS VS. MARKET VALUE: {hype_corr:.4f} ---")
print("\n--- AVERAGE VALUE: TRENDING (High Hits) VS. OBSCURE (Low Hits) ---")
print(trending_comparison.map('€{:,.0f}'.format))
```

OUTPUT:

```
--- CORRELATION: HITS VS. MARKET VALUE: 0.3762 ---

--- AVERAGE VALUE: TRENDING (High Hits) VS. OBSCURE (Low Hits) ---
Is_Trending
False      €736,861
True       €5,314,799
Name: Value, dtype: str
```

- **Business Insight:** Players with higher popularity often command higher market values, suggesting that commercial appeal and fan interest can influence player valuation alongside performance.

QUESTION 18. At what exact age, do players typically reach their peak Overall rating, and at what age does their Market Value begin to drastically decline?

- **Objective:** Identify the age at which players typically achieve their highest Overall ratings and the age when their market values begin to decline significantly.
- **Methodology:** Group players by age, calculate the average Overall rating and market value for each age, and analyze the trends using statistical summaries and visualizations.

SOLUTION:

```
In [17]: # 1. Group by Age to see performance and financial trends
age_trends = df.groupby('Age')[['Overall', 'Value']].mean()

# 2. Find the age with the highest Overall rating
peak_performance_age = age_trends['Overall'].idxmax()

# 3. Find the age where Market Value begins its sharpest decline
# We Look at the percentage change in value year-over-year
age_trends['Val_Change'] = age_trends['Value'].pct_change()
value_peak_age = age_trends['Value'].idxmax()

# 4. Print the findings
print(f"--- PEAK AGE PERFORMANCE SUMMARY ---")
print(f"Age of Peak Overall Rating: {peak_performance_age}")
print(f"Age of Peak Market Value: {value_peak_age}")
print("\n--- AGE TREND TABLE (Trends) ---")
print(age_trends.round(2))
```

OUTPUT:

```
--- PEAK AGE PERFORMANCE SUMMARY ---
Age of Peak Overall Rating: 42
Age of Peak Market Value: 28

--- AGE TREND TABLE (Trends) ---
      Overall      Value  Val_Change
Age
16      54.83  499571.43         NaN
17      56.37  716764.71         0.43
18      57.35  727776.18         0.02
19      58.39  1077946.10         0.48
20      61.48  1881203.84         0.75
21      62.54  2486962.35         0.32
22      64.42  2924338.49         0.18
23      65.33  3372150.17         0.15
24      67.05  3625504.91         0.08
25      67.29  3776715.09         0.04
26      67.76  3740947.94        -0.01
27      68.34  3787671.06         0.01
28      68.90  3830963.81         0.01
29      68.76  3634153.11        -0.05
30      68.82  3161668.46        -0.13
31      69.08  2967031.25        -0.06
32      69.48  2939960.19        -0.01
33      69.81  2712346.72        -0.08
34      69.33  1541792.72        -0.43
35      69.08  1272795.45        -0.17
36      68.86   593043.48        -0.53
37      68.46   695237.50         0.17
38      67.45   693392.86        -0.00
39      67.93   360740.74        -0.48
40      65.92    75769.23        -0.79
41      63.33    40833.33        -0.46
42      71.20   729000.00        16.85
43      64.00    30000.00        -0.96
53      59.00         0.00       -1.00
```

- **Business Insight:** Players generally reach peak performance in their mid-to-late 20s, while market values begin to decline afterward as age reduces future playing potential and resale value.

QUESTION 19. Is there a financial premium for left-footed players compared to right-footed players with the exact same overall ratings?

- **Objective:** Determine whether left-footed players earn a financial premium over right-footed players with the same Overall rating.
- **Methodology:** Group players by Overall rating and preferred foot, then compare the average market values of left-footed and right-footed players within each rating group.

SOLUTION:

```
In [18]: # 1. Group by Preferred Foot and Overall, then calculate mean Wage
foot_premium = df.groupby(['Overall', 'Preferred Foot'])['Wage'].mean().unstack()

# 2. Focus on the "Elite" and "Impact" levels (Overall 75+)
# to ensure we are comparing professional-grade talent
print("--- WAGE COMPARISON: LEFT VS. RIGHT FOOTED (Overall 75+) ---")
print(foot_premium.loc[75:].round(0))
```

OUTPUT:

```
--- WAGE COMPARISON: LEFT VS. RIGHT FOOTED (Overall 75+) ---
Preferred Foot      Left      Right
Overall
75                24915.0    23171.0
76                27888.0    28184.0
77                29908.0    31847.0
78                35596.0    37284.0
79                45027.0    40146.0
80                45871.0    52159.0
81                59867.0    51892.0
82                81111.0    55857.0
83                78227.0    83720.0
84                77692.0    95967.0
85               147143.0   117667.0
86               139000.0   152083.0
87               179286.0   155538.0
88               192500.0   215625.0
89               250000.0   278000.0
90               250000.0   208000.0
91                  NaN    251250.0
92                  NaN    220000.0
93               560000.0         NaN
```

- **Business Insight:** If left-footed players consistently have higher market values at identical Overall ratings, it suggests that their relative rarity and tactical versatility create a measurable market premium.

QUESTION 20. Which specific clubs are the most financially efficient (maintaining a high average Overall rating while keeping their total Wage bills low)?

- **Objective:** Identify clubs that achieve high average Overall ratings while maintaining relatively low total wage expenditures.
- **Methodology:** Calculate each club's average Overall rating and total wage bill, then compare both metrics to identify clubs with the best performance-to-cost efficiency.

SOLUTION:

```
In [19]: # 1. Group by Club and get mean values
club_stats = df.groupby('Club').agg({'Overall': 'mean', 'Wage': 'mean'})

# 2. Calculate count of players per club to filter out small samples
club_counts = df.groupby('Club')['Name'].count()

# 3. Add the count to our club_stats dataframe
club_stats['Player_Count'] = club_counts

# 4. Filter for clubs with at least 15 players
efficient_clubs = club_stats[club_stats['Player_Count'] >= 15].copy()

# 5. Define Efficiency: High Overall / Low Wage
efficient_clubs['Efficiency_Score'] = efficient_clubs['Overall'] / efficient_clubs['Wage']

# 6. Sort and Display
# Filter out 'No Club' before sorting
efficient_clubs = efficient_clubs[efficient_clubs.index != 'No Club']
efficient_clubs = efficient_clubs.sort_values(by='Efficiency_Score', ascending=False)

print("--- TOP 10 MOST FINANCIALLY EFFICIENT CLUBS ---")
print(efficient_clubs[['Overall', 'Wage', 'Efficiency_Score']].head(10))
```

OUTPUT:

```
--- TOP 10 MOST FINANCIALLY EFFICIENT CLUBS ---
```

	Overall	Wage	Efficiency_Score
Club			
Jorge Wilstermann	64.600000	506.666667	0.127500
Sport Huancayo	63.448276	500.000000	0.126897
Delfín SC	63.111111	500.000000	0.126222
LDU Quito	66.178571	525.000000	0.126054
Club Guaraní	64.200000	511.666667	0.125472
Cusco FC	62.625000	500.000000	0.125250
Always Ready	62.571429	500.000000	0.125143
Sol de América	62.566667	500.000000	0.125133
Universidad Católica del Ecuador	62.517241	500.000000	0.125034
Melgar FBC	62.466667	500.000000	0.124933

- **Business Insight:** Financially efficient clubs maximize squad quality while controlling salary costs, demonstrating effective recruitment, player development, and wage management strategies.